Music Genre Classification: Enhancing Deep Learning and Traditional Machine Learning Approaches

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***Abstract*—The automatic classification of music genres is a highly significant problem in the field of audio processing. Its importance stems from two primary factors: the reduction of the manual burden associated with labeling massive audio databases, and the determination of the most significant spectral and statistical features that differentiate distinct music types. This project tackles the classification problem by building upon an established baseline that utilized both Traditional Machine Learning (ML) and Convolutional Neural Network (CNN) models on the GTZAN dataset. By developing a richer dataset representation comprising 95 spectral features for ML models, and introducing an overlapped chunking and normalized preprocessing methodology for CNN models, this study demonstrates a marked improvement in classification performance. Most notably, the proposed CNN architecture leverages the spatial features of Mel-spectrograms to achieve an accuracy of over 85%, significantly outperforming the baseline results.**

Keywords—Music Genre Classification, Machine Learning, Deep Learning, Convolutional Neural Networks (CNN), GTZAN Dataset, Mel-Spectrogram, Audio Signal Processing, Feature Extraction

I. Introduction

Associating a specific genre class with a given music track requires extracting distinguishing features from audio excerpts and using them to predict the correct category. As the volume of digital music continues to expand, robust machine learning and deep learning models are essential to automate categorization.

The baseline for this research is the 2021 study "*Music Genre Classification: A Review of Deep-Learning and Traditional Machine-Learning Approaches*" by Ndou et al. [1]. While the baseline provided a solid comparative framework using the GTZAN dataset, the model performances left room for improvement—particularly the CNN model, which achieved an accuracy of approximately 71%. The objective of this project is to enhance the existing body of work through superior dataset formation, refined feature selection, and optimized deep learning architectures.

II. Methodology

A. Baseline Reference

The baseline approach utilized the GTZAN dataset, which consists of 100 audio excerpts of 30-second duration for each of the 10 music genres. The original implementation applied direct division without overlaps for CNN inputs and evaluated both 3-second and 30-second duration features for traditional ML models. The CNN implementation in the baseline yielded an accuracy of around 71%.

B. Proposed Contributions

To improve upon the baseline, significant modifications were made to both the dataset generation methodology and the model-building stages.

Traditional Machine Learning Models: Instead of relying on basic feature extraction, a richer feature set was constructed for the traditional ML pipeline. Each 30-second excerpt was divided into 3-second chunks. From these chunks, a comprehensive set of 95 spectral and statistical features (including MFCCs, chroma, and tempo data) was generated. This enhanced dataset was then used to train and evaluate various models, including Random Forest Classifiers and Multi-Layer Perceptrons, adding to the k-Nearest Neighbors and Support Vector Machines explored in the baseline.

CNN Mel-Spectrogram Models: For the CNN dataset generation, an overlapped chunking strategy was implemented to create more meaningful samples of temporal information. The process was executed as follows:

- Each 30-second audio excerpt was processed into 19 distinct samples using 3-second chunks with a 1.5-second overlap.
- Mel-spectrograms were calculated for each sample using 128 Mel bands and a hop length of 512, creating a robust two-dimensional representation.
- To ensure data normalization and help the models converge properly during training, the raw power spectrograms were explicitly converted to decibel (dB) units (power_to_db) before being fed into the network.
- This generated a final dataset with dimensions of (19000, 130, 128).

A 2D Convolutional Neural Network architecture was employed to exploit the spatial features of this spectrogram dataset. To prevent the overfitting issues commonly encountered in such tasks, Dropout regularization and Batch Normalization layers were systematically incorporated into the network.

III. RESULTS

The modifications made to the feature extraction and network architecture yielded substantial improvements across the board.

TABLE I. Summary of Traditional ML Model Accuracies compared against baseline

Machine Learning Model	Baseline Accuracy (Ndou et al.)	Proposed Approach Accuracy (95 Features)	Absolute Improvement
k-Nearest Neighbors (kNN)	69.7%	84%	+14.3%
Support Vector Machine (SVM)	80.8%	85%	+4.2%
Random Forest (RF)	72.4%	85%	+12.6%
Multi-Layer Perceptron (MLP)	77.3%	85%	+7.7%

For the deep learning approach, the CNN trained on the overlapping Mel-spectrograms demonstrated vastly superior performance. Evaluated on the test set, the model achieved a categorical accuracy of 85.26% with a test set loss of 1.558.

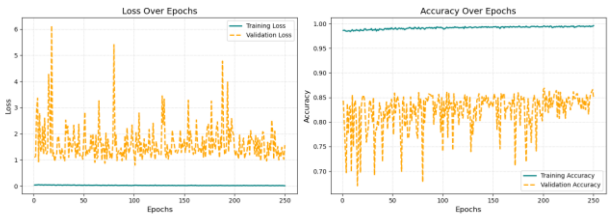

Fig. 1. CNN Training vs. Validation Accuracy and Loss Curves

When directly compared to the baseline article's CNN model, which stalled at roughly 71% accuracy, the proposed methodology secured an absolute improvement of over 14%. It is evident that the overlapping dataset generation, combined with proper decibel scaling, allows the network to represent and learn audio features much more robustly than direct track division.

IV. CONCLUSION

This project successfully expanded upon existing research in automatic music genre classification. By focusing on high-quality data representation—specifically through the extraction of 95 granular spectral features for traditional ML models and the use of overlapped, dB-scaled Mel-spectrograms for deep learning—the predictive capabilities of the algorithms were significantly enhanced. The integration of regularization techniques such as Dropout and Batch Normalization in the CNN architecture further stabilized training, ultimately pushing the classification accuracy to 85.26%. These results highlight that refined feature engineering and robust preprocessing are just as critical as model selection in complex audio classification tasks.

The implementation details and source code are publicly available at: https://github.com/keremtemel0/Music-Genre-Classification-GTZAN

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